



US012413793B2

(12) **United States Patent**
Fang et al.

(10) **Patent No.: US 12,413,793 B2**

(45) **Date of Patent: *Sep. 9, 2025**

(54) **GEOLOCATIONING SYSTEM AND METHOD FOR USE OF SAME**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 118 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/459,700**

(22) Filed: **Sep. 1, 2023**

(65) **Prior Publication Data**

US 2025/0076450 A1 Mar. 6, 2025

Related U.S. Application Data

(63) Continuation of application No. 17/752,146, filed on May 24, 2022, now Pat. No. 11,750,850, which is a (Continued)

(51) **Int. Cl.**
H04N 21/214 (2011.01)
G01S 5/08 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04N 21/2143** (2013.01); **G01S 5/08** (2013.01); **G06T 11/60** (2013.01); **G07C 1/00** (2013.01); **G08B 25/10** (2013.01); **H04N 21/23605** (2013.01); **H04N 21/25841**

(58) Field of Classification Search

CPC H04N 21/4516; H04N 21/43637; H04N 21/2143; H04N 21/25841; H04N 21/4343; H04N 21/4524; H04N 21/4108; H04N 21/414; H04N 21/42221; H04N 21/23605; G07C 1/00; H04W 84/12

See application file for complete search history.

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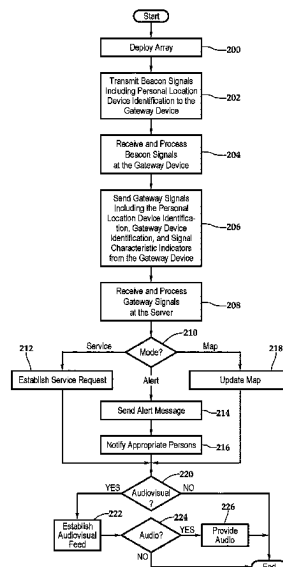
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(57) ABSTRACT

A geolocationing system and method for providing awareness in a multi-space environment, such as a hospitality environment or educational environment, are presented. In one embodiment of the geolocationing system, a vertical and horizontal array of gateway devices is provided. Each gateway device includes a gateway device identification providing an accurately-known fixed location within the multi-space environment. Each gateway device includes a wireless transceiver that receives a beacon signal from a proximate wireless-enabled personal locator device. The gateway devices, in turn, send gateway signals to a server, which determines estimated location of the wireless-enabled personal locator device.

20 Claims, 5 Drawing Sheets



Related U.S. Application Data

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continuation of application No. 17/154,713, filed on Jan. 21, 2021, now Pat. No. 11,343,543, which is a continuation of application No. 16/733,041, filed on Jan. 2, 2020, now Pat. No. 10,904,582, which is a continuation-in-part of application No. 16/201,783, filed on Nov. 27, 2018, now Pat. No. 10,602,196, which is a continuation of application No. 15/652,622, filed on Jul. 18, 2017, now Pat. No. 10,142,662, which is a continuation of application No. 15/165,851, filed on May 26, 2016, now Pat. No. 9,712,872, which is a continuation of application No. 14/461,479, filed on Aug. 18, 2014, now Pat. No. 9,357,254.

- (60) Provisional application No. 62/787,785, filed on Jan. 3, 2019, provisional application No. 61/935,862, filed on Feb. 5, 2014.

(51) **Int. Cl.**

G06T 11/60 (2006.01)
G07C 1/00 (2006.01)
G08B 25/10 (2006.01)
H04N 21/236 (2011.01)
H04N 21/258 (2011.01)
H04N 21/41 (2011.01)
H04N 21/414 (2011.01)
H04N 21/422 (2011.01)
H04N 21/434 (2011.01)
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H04N 23/66 (2023.01)
H04W 84/12 (2009.01)

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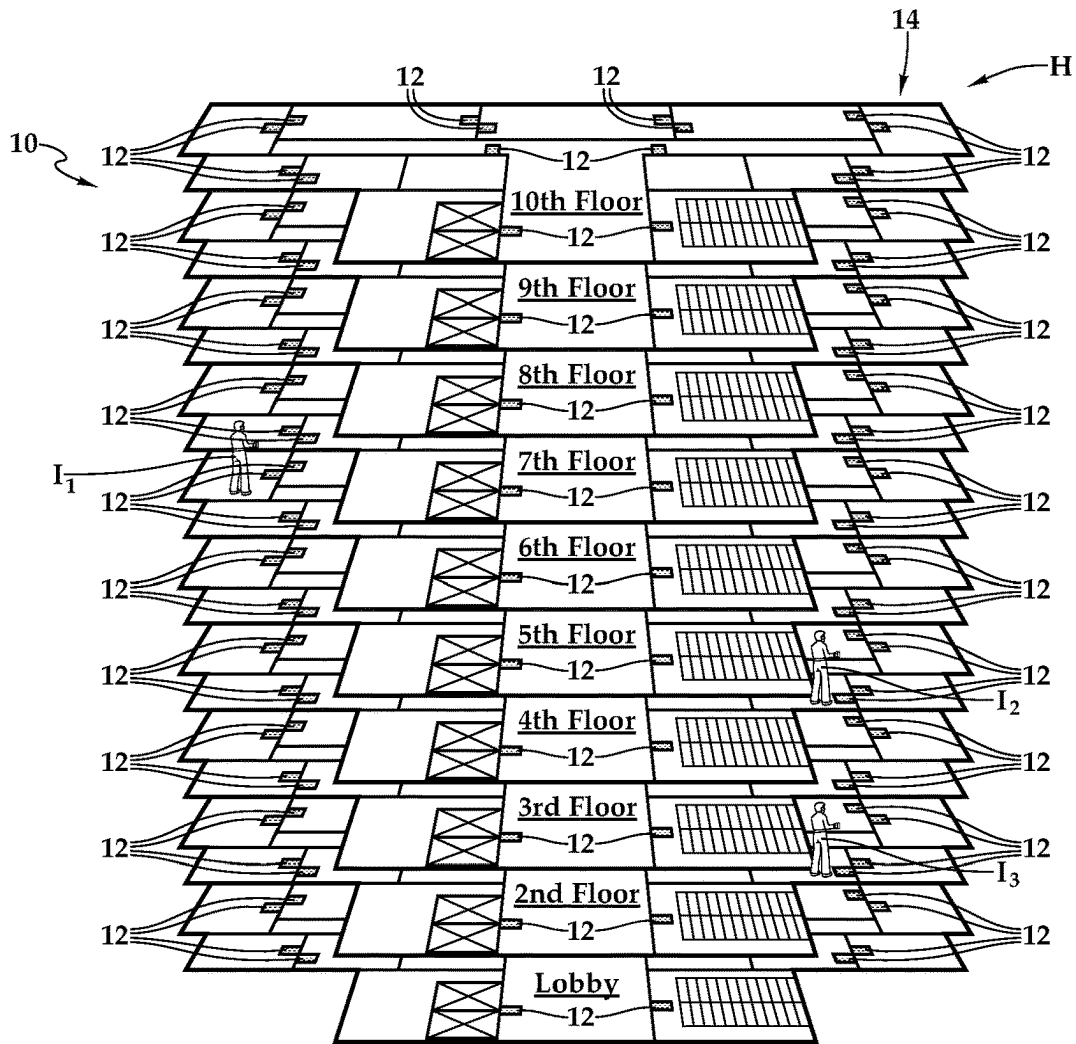


Fig.1A

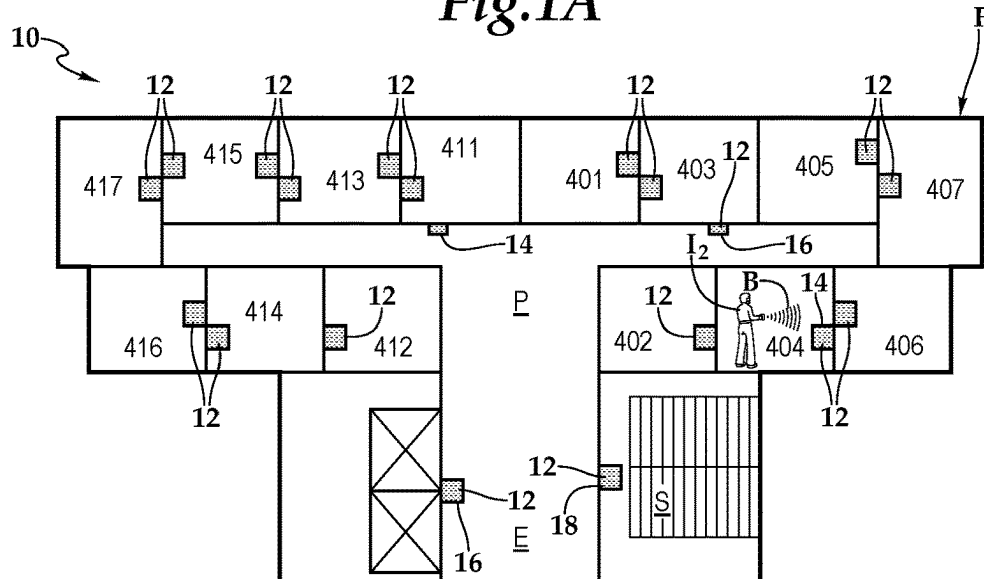


Fig.1B

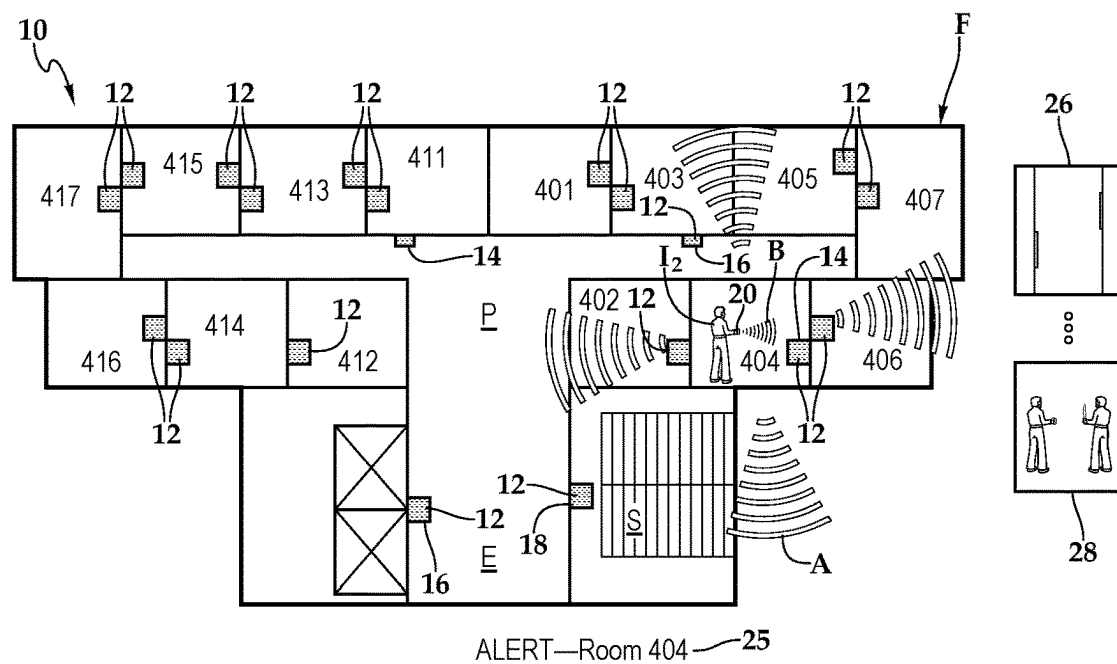
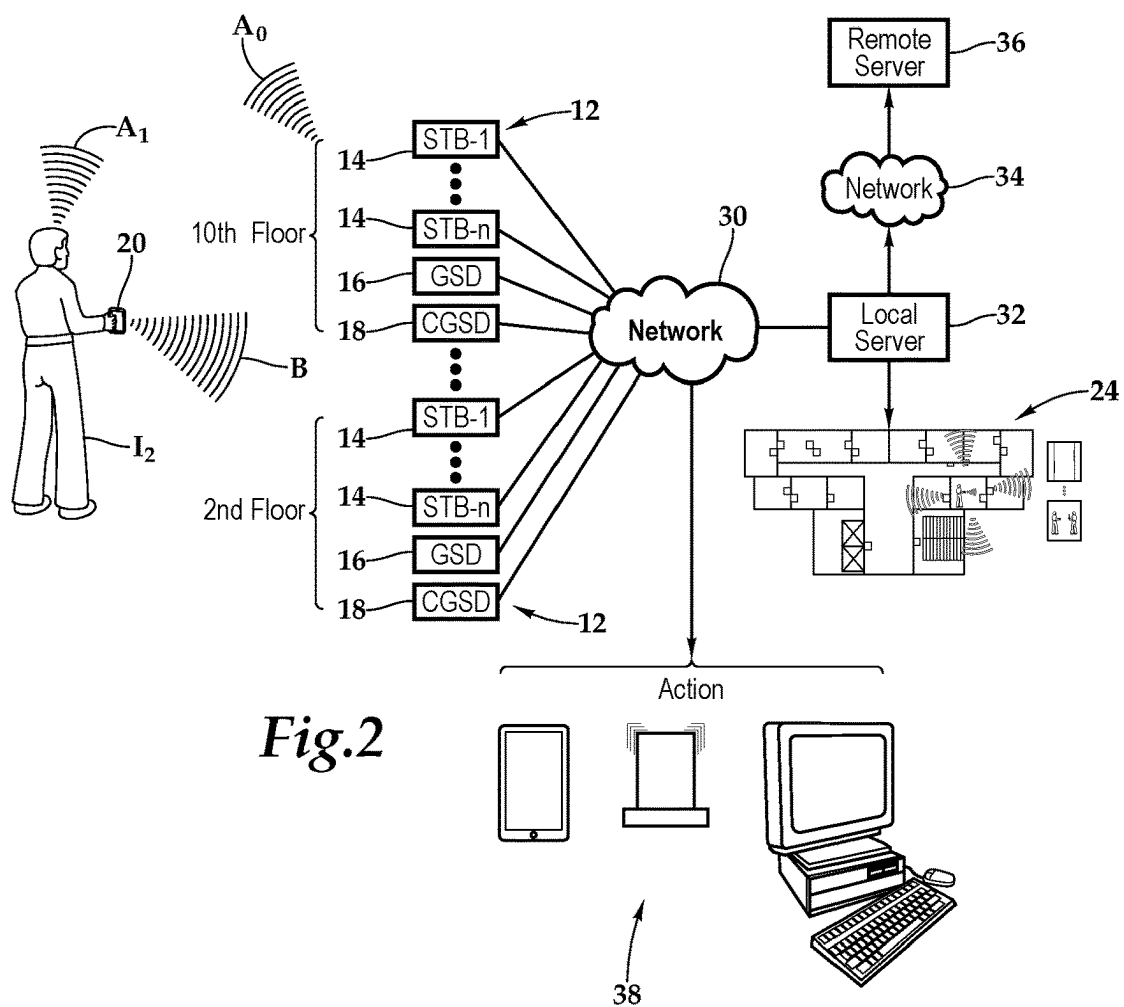
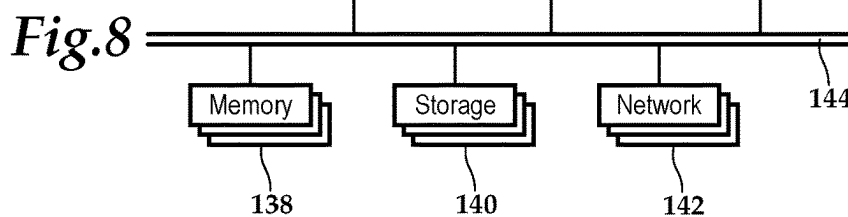
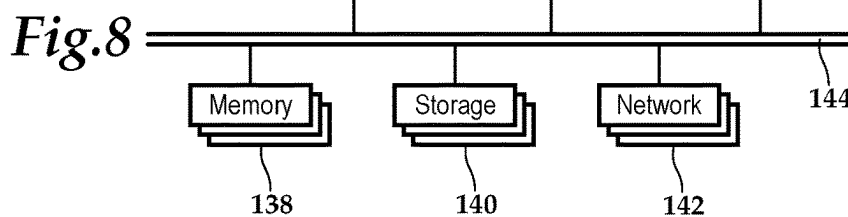
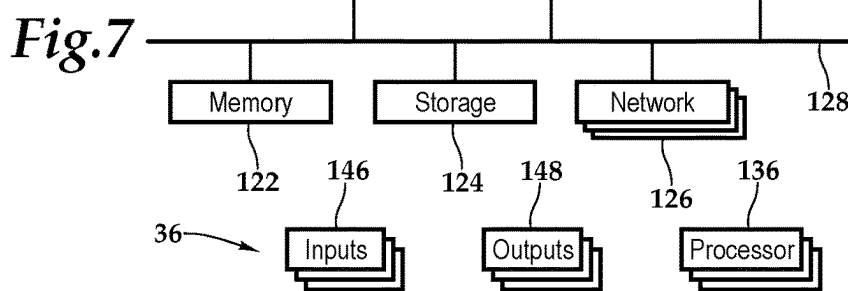
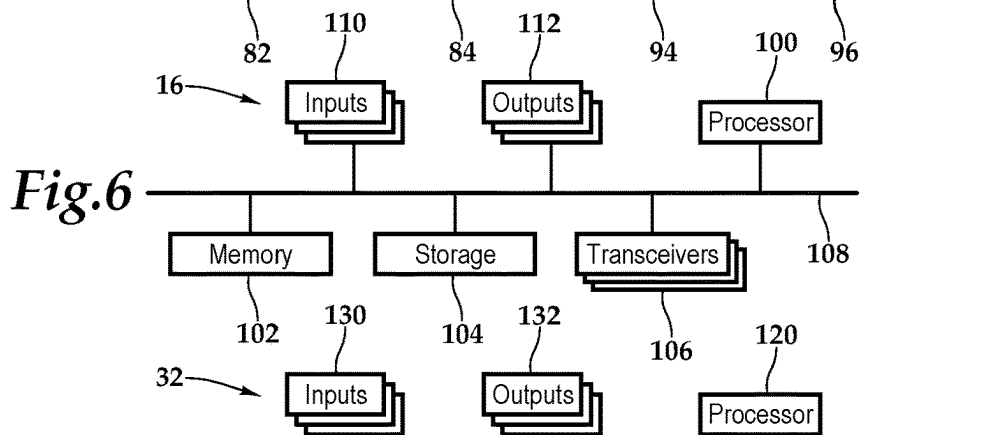
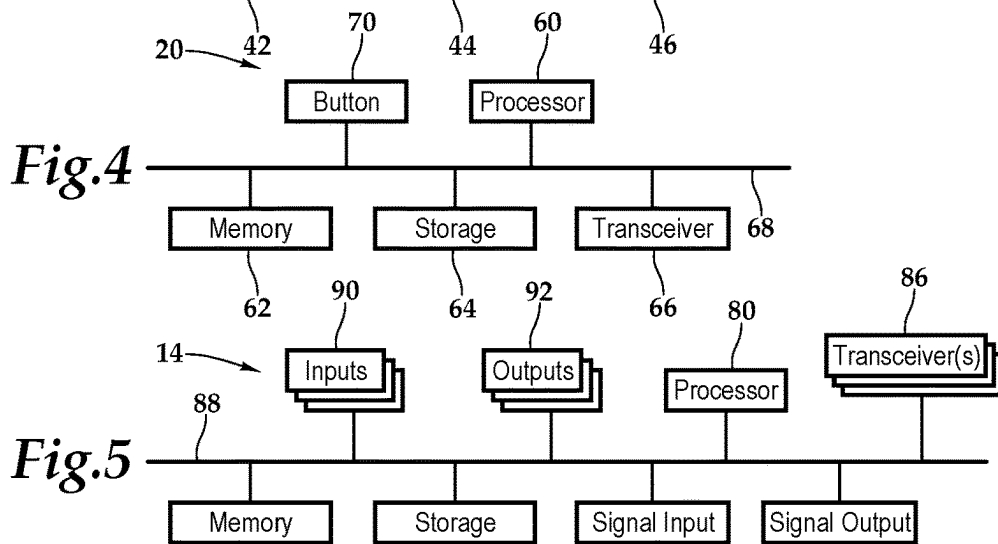
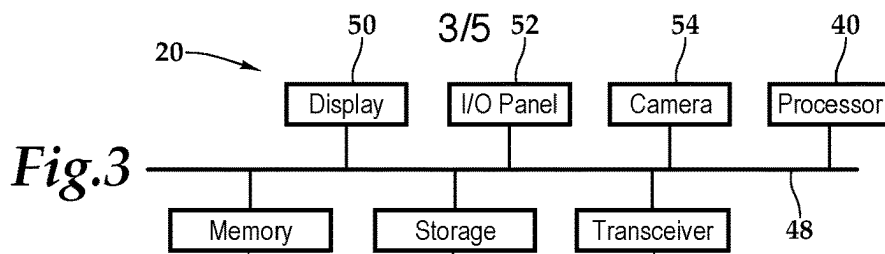


Fig.1C





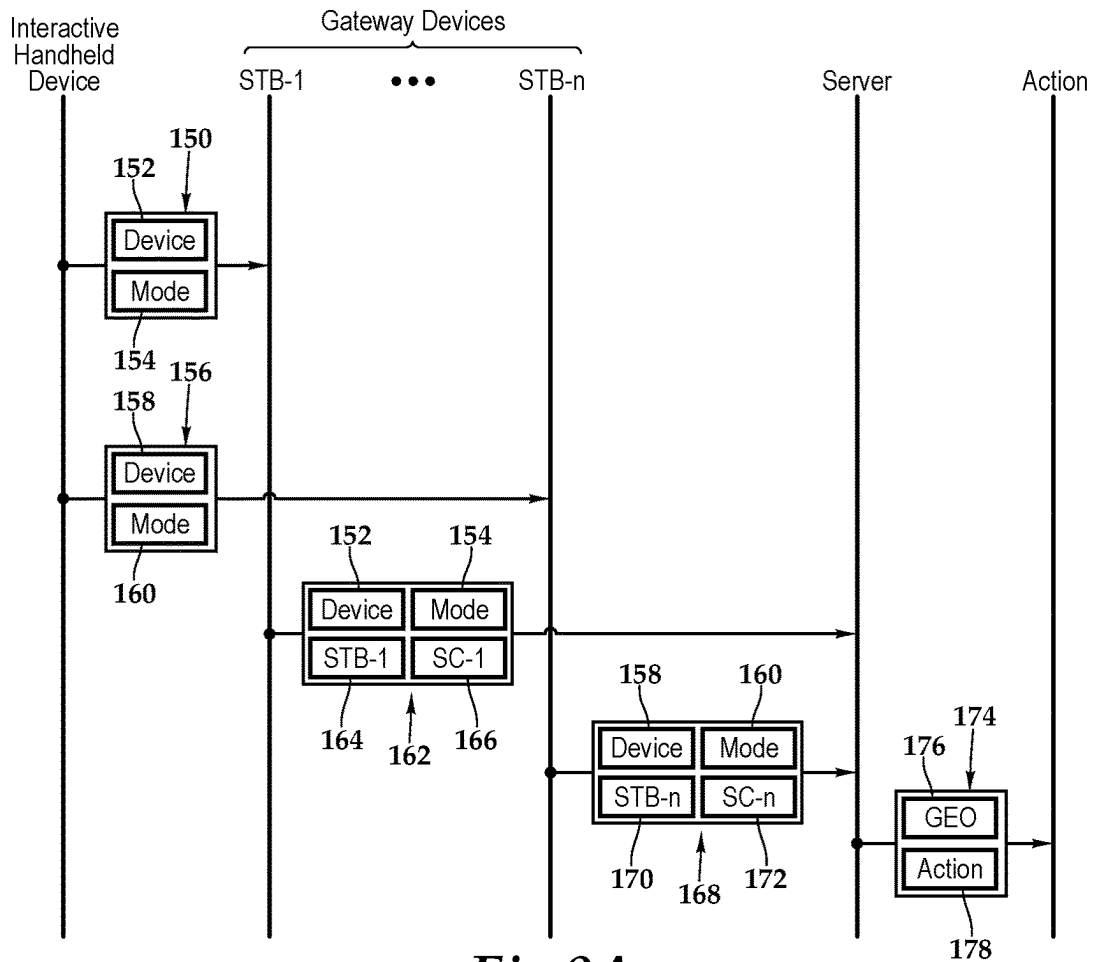


Fig.9A

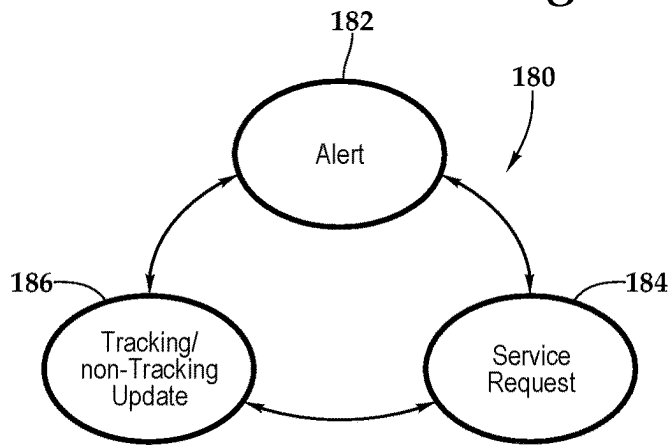


Fig.9B

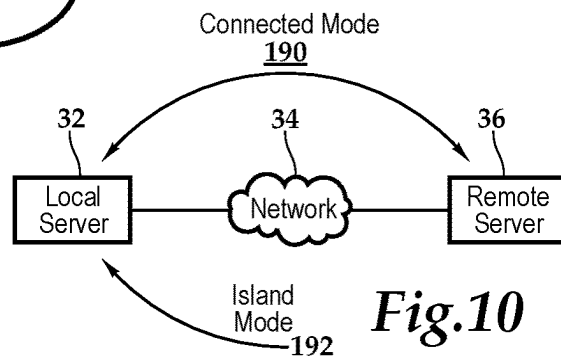
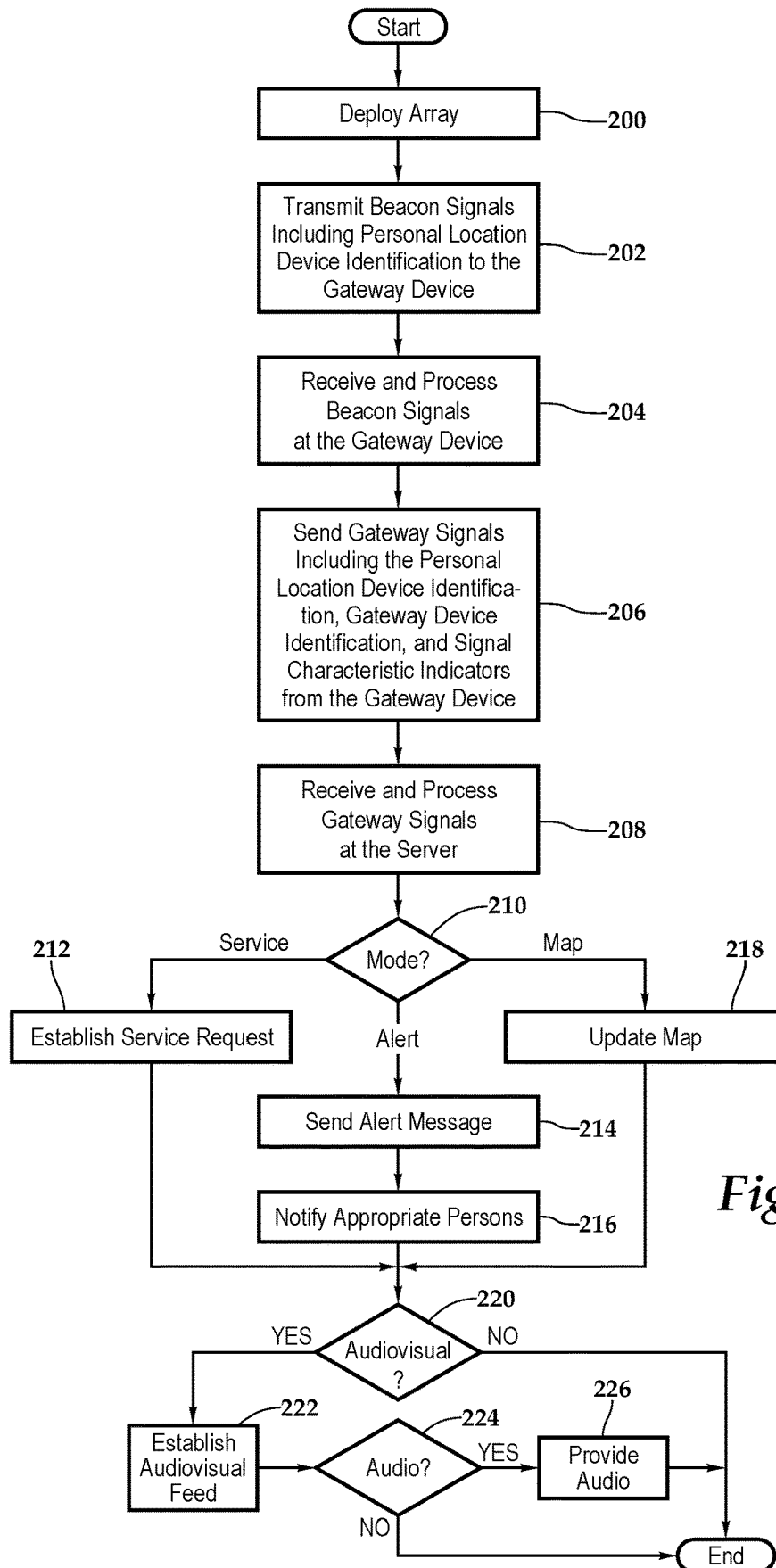


Fig.10

**Fig.11**

**GEOLOCATIONING SYSTEM AND METHOD
FOR USE OF SAME****PRIORITY STATEMENT & CROSS-REFERENCE
TO RELATED APPLICATION**

This application is a continuation of U.S. application Ser. No. 17/752,146 entitled "Geolocationing System and Method for Use of Same", filed on May 24, 2022 in the names of William C. Fang et al.; which is a continuation of U.S. application Ser. No. 17/154,713 entitled "Geolocationing System and Method for Use of Same", filed on Jan. 21, 2021 in the names of William C. Fang et al., now U.S. Pat. No. 11,343,543 issued on May 24, 2022; which is a continuation of U.S. application Ser. No. 16/733,041 entitled "Geolocationing System and Method for Use of Same," filed on Jan. 2, 2020 in the names of William C. Fang, et al., now U.S. Pat. No. 10,904,582 issued on Jan. 26, 2021; which claims priority from U.S. Patent Application Ser. No. 62/787,785 entitled "Geolocationing System and Method for Use of Same" filed on Jan. 3, 2019, in the name of William C. Fang; both of which are hereby incorporated by reference, in entirety, for all purposes. U.S. application Ser. No. 16/733,041 is also a continuation-in-part of U.S. patent application Ser. No. 16/201,783 entitled "Set-Top Box, System and Method for Providing Awareness in a Hospitality Environment" filed on Nov. 27, 2018, in the names of Vanessa Ogle et al., now U.S. Pat. No. 10,602,196 issued on Mar. 24, 2020; which is a continuation of U.S. patent application Ser. No. 15/652,622 entitled "Set-Top Box, System and Method for Providing Awareness in a Hospitality Environment" filed on Jul. 18, 2017, in the names of Vanessa Ogle et al., now U.S. Pat. No. 10,142,662 issued on Nov. 27, 2018; which is a continuation of U.S. patent application Ser. No. 15/165,851 entitled "Set-Top Box, System and Method for Providing Awareness in a Hospitality Environment" filed on May 26, 2016, in the names of Vanessa Ogle et al., now U.S. Pat. No. 9,712,872 issued on Jul. 18, 2017; which is a continuation of U.S. patent application Ser. No. 14/461,479 entitled "Set-Top Box, System and Method for Providing Awareness in a Hospitality Environment" filed on Aug. 18, 2014, in the names of Vanessa Ogle et al., now U.S. Pat. No. 9,357,254 issued on May 31, 2016; which claims priority from U.S. Patent Application Ser. No. 61/935,862 entitled "System and Method for Providing Awareness in a Hospitality Environment" and filed on Feb. 5, 2014, in the name of Vanessa Ogle; all of which are hereby incorporated by reference, in entirety, for all purposes.

TECHNICAL FIELD OF THE INVENTION

This invention relates, in general, to geolocationing and, in particular, to enhanced performance in systems and methods for providing awareness and safety in a multi-room environment such as a hospitality environment, educational environment, or the like.

BACKGROUND OF THE INVENTION

Without limiting the scope of the present invention, the background will be described in relation to employee safety in hospitality environments, as an example. Employees face increased personal security risks at work in multi-room environments such as hospitality environments, which include motels, hotels, and the like, for example. Such hospitality industry employees often work alone and range

over large interior areas that may be divided into many small, closed spaces. As a result of limited existing security measures, there is a need for improved systems and methods of providing awareness and safety in hospitality environments.

SUMMARY OF THE INVENTION

It would be advantageous to achieve systems and methods for providing geolocationing in a multi-room environment such as a hospitality environment, educational environment, or the like that would improve upon existing limitations in functionality. It would be desirable to enable an electrical engineering-based and software solution that would provide enhanced awareness and safety in an easy-to-use platform in the hospitality lodging industry or in another environment. To better address one or more of these concerns, a geolocationing system and method for use of the same are disclosed.

In one embodiment of the geolocationing system, a vertical and horizontal array of gateway devices is provided. Each gateway device includes a gateway device identification providing an accurately-known fixed location within the multi-space environment. Each gateway device includes a wireless transceiver that receives a beacon signal from a proximate wireless-enabled personal locator device. The gateway devices, in turn, send gateway signals to a server, which determine estimated location of the wireless-enabled personal location. These and other aspects of the invention will be apparent from and elucidated with reference to the embodiments described hereinafter.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of the features and advantages of the present invention, reference is now made to the detailed description of the invention along with the accompanying figures in which corresponding numerals in the different figures refer to corresponding parts and in which:

FIG. 1A is schematic building diagram depicting one embodiment of a geolocationing system for providing awareness in a multi-room environment illustrated as a hotel, according to the teachings presented herein;

FIG. 1B is a schematic floor plan depicting a floor of the hotel presented in FIG. 1A in further detail;

FIG. 1C is a schematic floor plan depicting a floor of the hotel presented in FIG. 1A during an alert event;

FIG. 2 is a schematic diagram depicting one embodiment of the geolocationing system presented in FIG. 1A providing enhanced awareness and safety functionality therewith according to the teachings presented herein;

FIG. 3 is a functional block diagram depicting one embodiment of a personal location device depicted in FIG. 2 in further detail;

FIG. 4 is a functional block diagram depicting another embodiment of a personal location device depicted in FIG. 2 in further detail;

FIG. 5 is a functional block diagram depicting one embodiment of a gateway device, a set-top box, presented in FIG. 1A;

FIG. 6 is a functional block diagram depicting one embodiment of a gateway device, a gateway service device, presented in FIG. 1A;

FIG. 7 is a functional block diagram depicting one embodiment of a local server presented in FIG. 2;

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FIG. 8 is a functional block diagram depicting one embodiment of a remote server presented in FIG. 2;

FIG. 9A is a data processing diagram depicting one embodiment of the geolocationing system according to the teachings presented herein;

FIG. 9B is a schematic diagram depicting one embodiment of the geolocationing system presented in FIG. 8A;

FIG. 10 is a schematic diagram depicting one embodiment of the operational modes of the local server and remote server; and

FIG. 11 is a flow chart depicting one embodiment of a method for providing a gateway device furnishing enhanced safety according to the teachings presented herein.

DETAILED DESCRIPTION OF THE INVENTION

While the making and using of various embodiments of the present invention are discussed in detail below, it should be appreciated that the present invention provides many applicable inventive concepts, which can be embodied in a wide variety of specific contexts. The specific embodiments discussed herein are merely illustrative of specific ways to make and use the invention, and do not delimit the scope of the present invention.

Referring initially to FIGS. 1A, 1B, 1C and 2, therein is depicted a geolocationing system for providing awareness in a multi-space environment such as a hospitality environment, which may be embodied as a furnished multi-family residence, dormitory, lodging establishment, hotel, hospital, which is schematically illustrated and designated 10. The multi-space environment may also be a multi-unit environment such as an educational environment like a school or college campus, for example. More generally, the geolocationing system 10 and the teachings presented herein are applicable to any multi-space environment including hospitality environments, educational campuses, hospital campuses, office buildings, multi-unit dwellings, sport facilities and shopping malls, for example.

As shown, by way of example and not by way of limitation, the multi-space environment is depicted as a hotel H having a lobby and floors F, which are appropriately labeled the 2nd floor through the 10th floor. Further, by way of example, the 4th floor is depicted with rooms 401, 402, 403, 404, 405, 406, 407, 411, 412, 413, 414, 415, 416, and 417. Additionally, a common area near the elevators is labeled E, a hallway labeled P, and a stairwell is labeled S. The lobby, the common area E, the hallway P, and the stairwell S are further illustrations of spaces in the multi-space environment in addition to the rooms.

Gateway devices 12 are deployed as part of a horizontal and vertical array, which is generally a spatial array, throughout the hotel H. It should be appreciated, however, that the gateway devices 12 and more generally deployment of the system 10 may include a horizontal array. Further, the deployment may be in a single story, multiple stories, or a combination thereof. As will be discussed in further detail hereinbelow, the gateway devices 12 may include set-top boxes 14, gateway service devices 16, or common space gateway devices 18.

Individuals, such as I₁, I₂, I₃, carry proximate wireless-enabled personal locator devices 20 which periodically, or on demand, transmit beacons that are received by a gateway device 12. The proximate wireless-enabled personal locator devices 20 may be a single button personal locator device or a proximate wireless-enabled interactive programmable device, such as a smart watch, a smart phone, or a tablet

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computer, for example. In one embodiment, the proximate wireless-enabled interactive programmable device 20 may be a wireless-enabled smart and interactive handheld device that may be supplied or carried by the user or guest. As shown individual I₁ works in the hospitality industry at hotel H and is presently working on the 4th floor. As the individual I₁ is working in room 404, the personal locator device 20 is transmitting beacons B that are received by gateway devices 12, such as the set-top box 14 that is located within the room 404 and the gateway service device 16 located in hallway P on the 4th floor of the hotel H.

As shown, the gateway device 12 in the room 404 is a set-top box 14, which may be connected to an electronic visual display device such as a display or television. The set-top box 14 may be an information appliance device that generally contains a TV-tuner as well as content input and display outputs. The set-top box 14 may be communicatively disposed with various amenities associated with the multi-space environment H as well as the system 10 providing a geolocation and safety network. The gateway device 12 in the hallway P of the 4th floor is a gateway service device 16 and a common space gateway device 18 is also in the hallway of the 4th floor. The gateway service device 16 may be communicatively disposed with various amenities associated with the multi-space environment H as well as the providing the geolocation and safety network. The common space gateway device 18 may include a limited set of functionalities as compared to the gateway service device 16. The limited functionality, however, includes connectivity to the system 10 providing the geolocation and safety network. Gateway devices, like the gateway device 12, including the set-top box 14, the gateway service device 16, and the common space gateway device 18 may be deployed throughout the spaces, and other areas of the hotel H or multi-space environment.

As mentioned, each of the gateway devices 12, including the set-top boxes 14, the gateway service devices 16, and the common space gateway devices 18, have a data link via a network 30 to a server 32 which is providing a geolocation and safety network. In one implementation, an individual I₂ has the proximate wireless-enabled personal locator device 20, which may transmit a beacon signal B from the personal locator device 20 using a wireless standard such as Wi-Fi to the gateway devices 12. Each of the gateway devices 12, including the set-top box 14 and the gateway service device 16, then processes the received beacon signal B and sends a gateway signal to the local server 32 or the remote server 36 by way of networks 30, 34. Under normal conditions, the beacon signals B and the gateway signals are sent to the remote server 36, which may be a cloud-based server. In this embodiment, the local server 32 may act as a monitoring station to notify an operator about the triggered alert and informing the operator about the alert condition. However, in the event of no connection to the remote server 36, such as during a period of time with no internet connectivity, the local server 32 assumes the responsibilities of the remote server 36. For purposes of illustration, the current embodiment described will consider an operational remote server 36.

The remote server 36 receives the gateway signals and uses multiple gateway signals for determining the estimated location of the proximate wireless-enabled personal locator device 20 of the individual I₂. The remote server 36, in turn, sends out the appropriate notifications to various phones, activates alarms, or notify others via a computer, depending on the situation, as shown by element 38. As a spatial array of horizontal and vertical gateway devices 12 are provided,

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the remote server 36 and system 10 presented herein is able to determine the location of the individual associated with the proximate wireless-enabled personal locator device 20 within a building. As particularly illustrated in FIGS. 1C and 2, the individual I_2 is in need of emergency assistance and activates the proximate wireless-enabled personal location device 20. In one implementation, beacon signals B are received by all nearby gateway devices 12, which in turn forward gateway signals to the remote server 36 for processing and determining the estimated location. The estimated location includes which floor F the individual is presently located as well as the room or common area E and the presence of a status or an alarm, such as Alarm A. In one embodiment, this information may be generated by the remote server 36 (or the local server 32) in the form of a map view 24, which includes a graphical representation of the multi-space environment that is annotated with the estimated location of the proximate wireless-enabled personal locator device 20. Further, the map view 24 includes an indication of the space in the form of an identification (e.g., Room 404) and status 25 (e.g., Alert) as well as one or more video feeds 26, 28 provided by cameras that are identified near the estimated location of the proximate wireless-enabled personal locator device 20. The map view 24 may be updated as the proximate wireless-enabled personal locator device 20 moves and corresponding audio and visual communications need to be adjusted too. In the illustrated example, a camera in the hallway and a camera within the Room 404 are activated. Further, as shown by audio input AI and audio output AO, one or two-way audio communication may be established with a nearby gateway device or the proximate wireless-enabled personal locator device 20.

In the systems presented herein, the video and audio may be activated in response to an alert 25 being triggered. Once the system 10 identifies the estimated location of the alert 25, the audio and video feeds from near the estimated location may be displayed at the local server 32 or another location. Alternatively, in public locations, the audio and/or video feeds may be ON continuously.

Referring to FIG. 3, the proximate wireless-enabled interactive programmable device 20 may be a wireless communication device of the type including various fixed, mobile, and/or portable devices. To expand rather the limit the previous discussion of the programmable device 20, such devices may include, but are not limited to, cellular or mobile telephones, two-way radios, personal digital assistants, digital music players, Global Positioning System units, tablet computers, smartwatches, and so forth. The programmable device 20 may include a processor 40, memory 42, storage 44, and a transceiver 46 interconnected by a bus architecture 48 that also supports a display 50, I/O panel 52, and a camera 54. It should be appreciated that although a particular architecture is explained, other designs and layouts are within the teachings presented herein.

In operation, the teachings presented herein permit a proximate wireless-enabled interactive programmable device 20 such as a smart phone or simple transmitter to communicate with a set-top box 14 that is able to relay an alert 25 with location information to a server and security or other individuals needing to know about the emergency. In one operational embodiment being described, the proximate wireless-enabled interactive programmable device 20 may be "paired" on a temporary basis to the set top/back box on a room-by-room basis, whereby the pairing changes as the hospitality employee's location changes. As shown, the proximate wireless-enabled interactive programmable device 20 includes the memory 42 accessible to the proces-

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sor 40 and the memory 42 includes processor-executable instructions that, when executed, cause the processor 40 to send beacon signals B. The proximate wireless-enabled interactive programmable device 20 may on-demand or periodically transmit the beacon signal B including a data packet having the programmable device identification, as well as a mode of operation identification.

Referring to FIG. 4, with respect to the simplified proximate wireless-enabled interactive programmable device 20, a processor 60, memory 62, storage 64, and a transceiver 66 are supported by an interconnected busing architecture 68. An emergency button 70 provides the activation that triggers the alert 25. As shown, the proximate wireless-enabled interactive programmable device 20 includes the memory 62 accessible to the processor 60 and the memory 62 includes processor-executable instructions that, when executed, cause the processor 60 to send beacon signals B. The proximate wireless-enabled interactive programmable device 20 may on-demand or periodically transmit the beacon signal B including a data packet having the programmable device identification as well as a mode of operation identification. In one embodiment, responsive to the activation of the emergency button 70, the proximate wireless-enabled interactive programmable device 20 immediately transmits a beacon signal B including a data packet having the programmable device identification as well as a mode of operation identification, i.e., an emergency alert.

Referring to FIG. 5, as used herein, set-top boxes, back boxes and set-top/back boxes may be discussed as set-top boxes. By way of example, the set-top box 14 may be a set-top unit that is an information appliance device that generally contains set-top box functionality including having a television-tuner: displays output through a connection to a display or television set and an external source of signal, turning by way of tuning the source signal into content in a form that can then be displayed on the television screen or other display device. Such set-top boxes are used in cable television, satellite television, and over-the-air television systems, for example.

The set-top box 14 includes a processor 80, memory 82, storage 84, and one or more transceivers 86 interconnected by a bus architecture 88 within a mounting architecture that supports inputs 90 and outputs 92. It should be understood that the processor 80, the memory 82, the storage 84, the inputs 90, and the outputs 92 may be entirely contained within the housing or the housing-dongle combination. The processor 80 may process instructions for execution within the computing device, including instructions stored in the memory 82 or in storage 84. The memory 82 stores information within the computing device. In one implementation, the memory 82 is a volatile memory unit or units. In another implementation, the memory 82 is a non-volatile memory unit or units. Storage 84 provides capacity that is capable of providing mass storage for the set-top box 14. Various inputs 90 and outputs 92 provide connections to and from the computing device, wherein the inputs 90 are the signals or data received by the set-top box 14, and the outputs 92 are the signals or data sent from the set-top box 14. A television content signal input 94 and a television output 96 are also secured in the housing in order to receive content from a source and forward the content, including external content such as cable and satellite and pay-per-view (PPV) programming, to the display.

The one or more transceivers 86 are associated with the set-top box 14 and communicatively disposed with the bus architecture 88. As shown, the transceivers 86 may be internal, external, or a combination thereof to the housing.

Further, the transceivers **86** may be a transmitter/receiver, receiver, or an antenna for example. Communication between various devices and the set-top box **14** may be enabled by a variety of wireless methodologies employed by the transceivers **86**, including 802.11, 3G, 4G, Edge, WiFi, ZigBee, near field communications (NFC), Bluetooth low energy and Bluetooth, for example. Also, infrared (IR) may be utilized.

The memory **82** and storage **84** are accessible to the processor **80** and include processor-executable instructions that, when executed, cause the processor **80** to execute a series of operations. With respect to the processor-executable instructions, the processor is caused to receive and process a beacon signal including a personal location device identification. More particularly, the processor-executable instructions cause the processor **80** to receive a beacon signal B via the wireless transceiver from a proximate wireless-enabled personal locator device **20**. The processor-executable instructions then cause the processor **80** to measure received signal characteristic of the beacon signal. The instructions may then cause the processor **80** to generate a gateway signal including the personal location device identification, a gateway device identification, and signal characteristics indicator, including received signal characteristic. Finally, the instructions may cause the processor **80** to send the gateway signal to the server **32**.

Referring to FIG. 6, the gateway device **12** may be a set-top unit that is an information appliance device that does not include television-tuner functionality and generally contains convenience and safety functionality. The gateway service device **16** includes a processor **100**, memory **102**, storage **104**, and transceivers **106** interconnected by a bus architecture **108** within a mounting architecture that supports inputs **110** and outputs **112**. The processor **100** may process instructions for execution within the computing device, including instructions stored in the memory **102** or in storage **104**. The memory **102** stores information within the computing device. In one implementation, the memory **102** is a volatile memory unit or units. In another implementation, the memory **102** is a non-volatile memory unit or units. Storage **104** provides capacity that is capable of providing mass storage for the gateway device **12**. Various inputs **110** and outputs **112** provide connections to and from the computing device, wherein the inputs **110** are the signals or data received by the gateway device **12**, and the outputs **112** are the signals or data sent from the gateway device **12**.

One or more transceivers **106** may be associated with the gateway device **12** and communicatively disposed with the bus **108**. The transceivers **106** may be internal, external, or a combination thereof to the housing. Further, the transceivers **106** may be a transmitter/receiver, receiver, or an antenna for example. Communication between various amenities in the hotel room and the gateway device **12** may be enabled by a variety of wireless methodologies employed by the transceivers **106**, including 802.11, 802.15, 802.15.4, 3G, 4G, Edge, Wi-Fi, ZigBee, near field communications (NFC), Bluetooth low energy and Bluetooth, for example. Also, infrared (IR) may be utilized.

The memory **102** and storage **104** are accessible to the processor **100** and include processor-executable instructions that, when executed, cause the processor **100** to execute a series of operations. With respect to the processor-executable instructions, the processor **100** is caused to receive and process a beacon signal B including a personal location device identification. More particularly, the processor-executable instructions cause the processor **100** to receive a beacon signal B via the wireless transceiver from a proximate

wireless-enabled personal locator device **20**. The processor-executable instructions then cause the processor **100** to measure a received signal characteristic of the beacon signal B. The instructions may then cause the processor **100** to generate a gateway signal including the personal location device identification, a gateway device identification, and signal characteristics indicator. Finally, the instructions may cause the processor **100** to send the gateway signal to the server **32**.

Referring now to FIG. 7, one embodiment of the local server **32** as a computing device includes a processor **120**, memory **122**, storage **124**, and one or more network adapters **126** interconnected with various buses **128** in a common or distributed, for example, mounting architecture, that supports inputs **130** and outputs **132**. In other implementations, in the computing device, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Further still, in other implementations, multiple computing devices may be provided and operations distributed therebetween. The processor **120** may process instructions for execution within the local server **32**, including instructions stored in the memory **122** or in storage **124**. The memory **122** stores information within the computing device. In one implementation, the memory **122** is a volatile memory unit or units. In another implementation, the memory **122** is a non-volatile memory unit or units. Storage **124** includes capacity that is capable of providing mass storage for the local server **32**. The various inputs **130** and outputs **132** provide connections to and from the local server **32**, wherein the inputs **130** are the signals or data received by the local server **32**, and the outputs **132** are the signals or data sent from the local server **32**. The network adapters **126** couples the local server **32** to a network such that the local server **32** may be part of a network of computers, a local area network (LAN), a wide area network (WAN), an intranet, a network of networks, or the Internet, for example.

The memory **122** and storage **124** are accessible to the processor **120** and include processor-executable instructions that, when executed, cause the processor **120** to execute a series of operations. In one embodiment of processor-executable instructions, the processor-executable instructions cause the processor **120** to receive a plurality of gateway signals from a plurality of gateway devices of the vertical and horizontal array. The processor **120** is caused to process the plurality of gateway signals and determine estimated location of the proximate wireless-enabled personal locator device **20**. The processor **120** may also be caused to annotate the graphical representation of the multi-space environment with location of the proximate wireless-enabled personal locator device **20**, and annotate the graphical representation of the room with the alert notification.

Referring now to FIG. 8, one embodiment of the remote server **36** as a computing device includes a processor **136**, memory **138**, storage **140**, and one or more network adapters **142** interconnected with various buses **144** in a common or distributed, for example, mounting architecture, that supports inputs **146** and outputs **148**. In other implementations, in the computing device, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Further still, in other implementations, multiple computing devices may be provided and operations distributed therebetween. The processor **136** may process instructions for execution within the remote server **36**, including instructions stored in the memory **138** or in storage **140**. The memory **138** stores information within the computing device. In one implemen-

tation, the memory **138** is a volatile memory unit or units. In another implementation, the memory **138** is a non-volatile memory unit or units. Storage **140** includes capacity that is capable of providing mass storage for the remote server **36**. The various inputs **146** and outputs **148** provide connections to and from the remote server **36**, wherein the inputs **146** are the signals or data received by the remote server **36**, and the outputs **148** are the signals or data sent from the remote server **36**. The network adaptor **142** couples the remote server **36** to a network such that the remote server **36** may be part of a network of computers, a local area network (LAN), a wide area network (WAN), an intranet, a network of networks, or the Internet, for example.

The memory **138** and storage **140** are accessible to the processor **136** and include processor-executable instructions that, when executed, cause the processor **136** to execute a series of operations. In one embodiment of processor-executable instructions, the processor-executable instructions cause the processor **136** to receive a plurality of gateway signals from a plurality of gateway devices of the vertical and horizontal array. The processor **136** is caused to process the plurality of gateway signals and determine estimated location of the proximate wireless-enabled personal locator device **20**. The processor **136** may also be caused to annotate the graphical representation of the multi-space environment with location of the proximate wireless-enabled personal locator device **20**, and annotate the graphical representation of the room with the alert notification.

FIG. 9A illustrates one embodiment of signalization and data transfer. As shown, a proximate wireless-enabled interactive programmable device **20** transmits data packet **150**, which is a beacon signal, including a device indicator **152** and a mode of operation indicator **154**. The proximate wireless-enabled interactive programmable device **20** also transmits data packet **156**, which is a beacon signal, including a device indicator **158** and a mode of operation indicator **160**. The data packets **150**, **156** are received by gateway devices; namely, set-top box STB-1 and set-top box STB-n. The gateway device STB-1 then establishes data packet **162**, including device indicator operation indicator **154**, gateway device identification **164** (STB-1), and signal characteristic **166** (SC-1). Similarly, the gateway device STB-n then establishes data packet **168**, including device indicator **158**, mode of operation indicator **160**, gateway device identification **170** (STB-n), and signal characteristic **172** (SC-n).

The data packets **162**, **168**, which are gateway signals, are transmitted to the server and the server analyzes the data packets **162**, **168** and determines the estimated location of the proximate wireless-enabled interactive programmable device **20**. The server then sends out signal **174**, which includes the estimated geolocation **174** and the appropriate action **178**.

FIG. 9B depicts one embodiment of a state diagram **180** of the states of the system **10**, which include an alert mode of operation **182**, a service request mode of operation **184**, and a tracking/non-tracking update mode of operation **186**. As will be appreciated, the modes of operation may overlap or, to a partial or full extent be combined. In the alert mode of operation **182**, a user of a proximate wireless-enabled interactive programmable device **20** may send an alert to indicate distress. In the service request mode of operation **184**, the user may send a service along with the location information. The tracking/non-tracking update mode of operation **184** indicates the level of privacy the user expects and how much of the location history will be saved.

FIG. 10 depicts one embodiment of the operations of the local server **32** and the remote server **36** showing the

connected mode **190** and the island mode **192**. As discussed, under normal conditions, the beacon signals and the gateway signals are sent to the remote server **36**, which may be a cloud-based server. In this embodiment, the local server **32** may act as a monitoring station to notify an the operator about the triggered alert and informing operator about the alert condition. However, in the event of no connection to the remote server **36**, such as during a period of time with no internet connectivity, the local server **32** assumes the responsibilities of the remote server **36**. For purposes of illustration, the current embodiment will be described will consider an operational remote server **36**.

FIG. 11 depicts one embodiment of a method for providing safety in a hospitality environment or other environment, according to the teachings presented herein. At block **200**, the array of gateway devices is deployed vertically and horizontally throughout the hospitality environment. At block **202**, beacon signals are periodically transmitted from personal locator devices and received by the gateway devices.

At block **204**, the beacon signals are received and processed at the gateway device. The beacon signals may include a personal location device identification corresponding to the device being employed by the user. In one embodiment, signal strength between the beacon transmission of the set-top boxes and the common area beacons at the wireless-enabled interactive programmable device is measured. In other embodiments, phase angle measurements or flight time measurements may be utilized. At block **206**, broadcast signals are sent from the gateway devices to a server that is part of the geolocation and safety network. The broadcast signals may include the personal location device identification, gateway device identification, and signal characteristic indicators. At block **208**, the server receives and processes the broadcast signals to determine an estimated location. At decision block **210**, the server takes action based on the mode of operation. In a first mode of operation at block **212**, a service request is associated with the location of the user utilizing the location of the personal location device such as the wireless-enabled interactive programmable device as a proxy. In a second mode of operation at block **214**, an emergency alert is sent and subsequent notification (block **216**) occurs. The emergency alert includes an indication of distress and the location of the user utilizing the location of the wireless-enabled interactive programmable device as a proxy. In a third mode of operation at block **218**, the map of individuals is updated with the location of the user with, if privacy settings being enabled, the system maintains the privacy of the individual working in the hospitality environment such that the system only retains in memory the last known position and time of the user-supplied wireless-enabled smart and interactive programmable device. Further, in this mode of operation, the system does not reveal the location of the individual and programmable device unless and until an alert is issued.

The order of execution or performance of the methods and data flows illustrated and described herein is not essential, unless otherwise specified. That is, elements of the methods and data flows may be performed in any order, unless otherwise specified, and that the methods may include more or less elements than those disclosed herein. For example, it is contemplated that executing or performing a particular element before, contemporaneously with, or after another element are all possible sequences of execution.

While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and

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combinations of the illustrative embodiments as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is, therefore, intended that the appended claims encompass any such modifications or embodiments.

What is claimed is:

1. A method for providing awareness in a multi-space environment, the method being executed by a processor in communication with a server via a programming interface, the server having non-transitory memory and storage inter-connected in a busing architecture, the method comprising:

receiving, via an array of gateway devices each having a gateway device identification providing an accurately-known fixed location, a plurality of gateway signals, each gateway signal including a personal locator device identification associated with a proximate wireless-enabled personal location device and a gateway device identification associated with a transmitting gateway device;

processing the plurality of gateway signals; and
determining an estimated location of the proximate wireless-enabled personal location device based on the processed gateway signals.

2. The method as recited in claim 1, wherein the gateway device is a set-top box.

3. The method as recited in claim 1, wherein the gateway device is a common space gateway device.

4. The method as recited in claim 1, wherein the gateway device is a gateway service device.

5. The method as recited in claim 1, wherein the proximate wireless-enabled personal location device is a single button personal locator device.

6. The method as recited in claim 1, wherein the proximate wireless-enabled personal location device is an interactive programmable device.

7. The method as recited in claim 6, wherein the interactive programmable device is selected from the group consisting of smartwatches, smartphones, and tablet computers.

8. The method as recited in claim 1, wherein the server is a back-office hotel server.

9. The method as recited in claim 1, wherein the server is a cloud-based server.

10. The method as recited in claim 1, further comprising receiving a distress signal from the proximate wireless-enabled personal location device.

11. The method as recited in claim 1, further comprising activating a camera proximate to the estimated location.

12. The method as recited in claim 11, wherein the camera is part of a device selected from the group consisting of personal locator devices, cellular telephones, mobile telephones, two-way radios, personal digital assistants, digital music players, Global Position System units, tablet computers, and smartwatches.

13. The method as recited in claim 11, wherein the camera is part of a device selected from the group consisting of gateway devices, set-top boxes, gateway service devices, and common space gateway devices.

14. The method as recited in claim 11, further comprising:
rendering a map view of the multi-space environment;
and

annotating the map view with a location of the camera.

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15. A method for providing awareness in a multi-space environment, the method being executed by a processor in communication with a server via a programming interface, the server having non-transitory memory and storage inter-connected in a busing architecture, the method comprising:

receiving, via an array of gateway devices each having a gateway device identification providing an accurately-known fixed location, a plurality of gateway signals, each gateway signal including a personal locator device identification associated with a proximate wireless-enabled personal location device and a gateway device identification associated with a transmitting gateway device;

processing the plurality of gateway signals;
determining an estimated location of the proximate wireless-enabled personal location device based on the processed plurality of gateway signals;

rendering a map view of the multi-space environment, the map view including a graphical representation of the multi-space environment; and

annotating the graphical representation of the multi-space environment with the estimated location of the proximate wireless-enabled personal location device.

16. The method as recited in claim 15, wherein the gateway device is a set-top box.

17. The method as recited in claim 15, wherein the gateway device is a common space gateway device.

18. The method as recited in claim 15, wherein the gateway device is a gateway service device.

19. A method for providing awareness in a multi-space environment, the method being executed by a processor in communication with a server via a programming interface, the server having non-transitory memory and storage inter-connected in a busing architecture, the method comprising:

receiving, via an array of gateway devices each having a gateway device identification providing an accurately-known fixed location, a plurality of gateway signals, each gateway signal including a personal locator device identification associated with a proximate wireless-enabled personal location device and a gateway device identification associated with a transmitting gateway device, the gateway signals further including an alert notification;

processing the plurality of gateway signals;
determining an estimated location of the proximate wireless-enabled personal location device based on the processed plurality of gateway signals;

rendering a map view of the multi-space environment, the map view including a graphical representation of the multi-space environment;

annotating the graphical representation of the multi-space environment with the estimated location of the proximate wireless-enabled personal location device; and
annotating the graphical representation of the multi-space environment with the alert notification.

20. The method as recited in claim 19, wherein the server is selected from the group consisting of back-office hotel servers and cloud-based servers.

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